

Predictive Modeling and Early Detection of White Spot Disease in Shrimp Farming using Machine Learning: A Case Study in Bangladesh

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Abstract

This paper addresses the severe threat of White Spot Disease (WSD) to the shrimp farming industry in Bangladesh and proposes a comprehensive methodology for predictive modeling and early detection using advanced machine learning techniques. The study emphasizes the importance of shrimp aquaculture in Bangladesh, outlines the impact of WSD outbreaks, and sets objectives for developing a specific prediction model for shrimp farming. Employing Random Forest, Multinomial Naive Bayes, and Bagging with Decision Trees, the research achieves impressive accuracy rates exceeding 90%, with the Bagging with Decision Trees classifier and Random Forest reaching a maximum accuracy of 97.87%. The outcomes hold significant implications for early WSD identification in the shrimp farming industry, contributing to Bangladesh's GDP and providing valuable insights for managing aquatic disease outbreaks globally.

Keywords: Shrimp Farming, White Spot Disease (WSD), Aquaculture, Machine Learning, Economic Impact

1 Introduction

In Bangladesh, shrimp farming has long been a pillar of the country’s agricultural and economic structure. Bangladesh has become one of the top producers of shrimp in the world because of its large coastline and ideal climatic conditions [1]. Millions of people, especially in rural regions, have employment chances, greatly enhancing the nation’s GDP. However, a chronic and fatal threat known as White Spot Disease (WSD) has shadowed the development and profitability of Bangladesh’s prawn farming business [2].

The vital necessity for early WSD identification in shrimp farming is what matters. It is vital to recognize WSD early on to halt the disease’s course and avoid financial losses [2]. The shrimp farming sector is vulnerable to catastrophic outbreaks because standard detection approaches usually fall short in accuracy and timeliness.

WSD poses a severe threat to shrimp farms, spreading rapidly and causing significant financial losses within 3 to 10 days, ultimately affecting the entire shrimp industry [3]. Our research is motivated by the urgent need to address this critical challenge in shrimp farming. To tackle this issue, we have outlined the following key objectives: Firstly, we aim to develop a robust prediction model specifically tailored for shrimp farming, focusing on early detection of White Spot Disease. Secondly, we intend to leverage modern visualization techniques and advanced machine learning algorithms to enhance the accuracy and efficiency of disease diagnosis within the shrimp farming environment. Lastly, our study aims to explore the potential of technology-driven solutions in mitigating the adverse impacts of White Spot Disease on shrimp farming operations. This includes identifying innovative approaches to minimize its detrimental effects, emphasizing the importance of incorporating novel strategies [3, 4].

The paper is structured as follows: Section 2 delves into related works, Section 3 outlines the research methodology, Section 4 presents the experimental results and a concise discussion, and Section 5 concludes the study, highlighting future directions.

2 Literature Review

The white spot syndrome virus (WSSV) poses a significant threat to global shrimp farming due to its catastrophic effects on shrimp populations. In Bangladesh, where shrimp farming is economically vital, understanding and managing WSSV incidence are crucial. WSSV outbreaks lead to rapid mortality, causing substantial financial losses for shrimp producers and negatively impacting the local industry [2].

WSSV induces white spot disease (WSD) in shrimp, resulting in rapid mortality within 3 to 10 days. Since its initial outbreak in China and Taiwan in 1991 and 1992, WSD has caused substantial losses globally [5]. Shrimp farming faces challenges such as feed quality, production costs, and diseases like WSD, which can be catastrophic if not diagnosed early [6].

Transfer learning models have shown promise in diagnosing shrimp diseases, with Ahmed et al. proposing a strategy achieving outstanding validation accuracies using MobileNetV1 and Support Vector Machine (SVM) approaches [6]. To address worsening conditions for shrimp producers and limitations of existing models, an Ensemble

learning model has been constructed to forecast WSD, offering hope in controlling its spread [6].

In Vietnam, where shrimp from the Mekong Delta plays a crucial role in global shrimp production, there's a rising concern about illnesses in diverse shrimp farming zones. Transfer learning applied to deep convolutional neural networks achieved a 92% accuracy in classifying six prevalent shrimp illnesses [7].

Traditional methods for identifying WSD, such as histopathology and viral isolation, are time-consuming. Studies have explored the use of bootstrapping random forest and CHAID for WSD prediction. Akter et al. utilized bootstrapping random forest in a Bangladesh dataset, achieving high accuracy [8]. Hossain et al. used CHAID in a Vietnamese dataset, obtaining accuracy significantly higher than traditional methods [8]. The proposed methods offer efficient and accurate alternatives for WSD prediction.

3 Methodology

In this section, we outline the methodology utilized in this study. The proposed method encompasses several key steps: data collection, preprocessing, feature engineering, and model selection. Figure 1 provides a graphical representation of the proposed methodology.

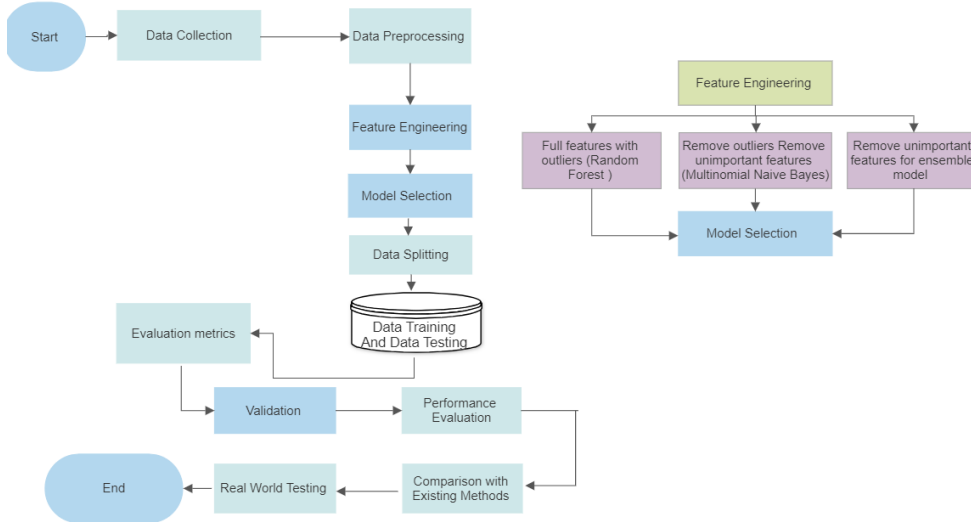


Fig. 1 Proposed shrimp white spot disease (WSD) detection methodology.

3.1 Data Collection and Preprocessing

The dataset employed in this study is collected from Mendeley data upon the reference of the study titled “Bootstrapping Random Forest and CHAID for WSD Prediction” [8, 9]. This dataset contains 48 features and 233 data points. The dataset includes

crucial features for shrimp farming, like pH level, current prevalence, water level, shrimp disease and other important factors. After collecting the dataset, missing values are handled with *fillna()* method using the mean values. After that, categorical values are converted to numerical values using one hot encoding method.

3.2 Feature Engineering

The feature engineering process involves selecting a subset of relevant and significant features (variables or characteristics) from the initial set of features in a dataset [10]. Our research initially assesses the performance of key models like Random Forest, Multinomial Naive Bayes and Logistic Regression by examining their feature importance graphs. Next, we eliminate less impactful features, leaving us with 23 crucial features related to shrimp production out of the initial 48. This streamlined approach makes our work smoother and more efficient. Feature selection helps simplify the model, decreases overfitting, and boosts its ability to generalize patterns from the data. Figure 2(a) shows the dataset's all features' correlation heatmap. In comparison, Figure 2(b) displays a more sophisticated representation that only emphasizes the significant elements chosen using a feature selection approach.

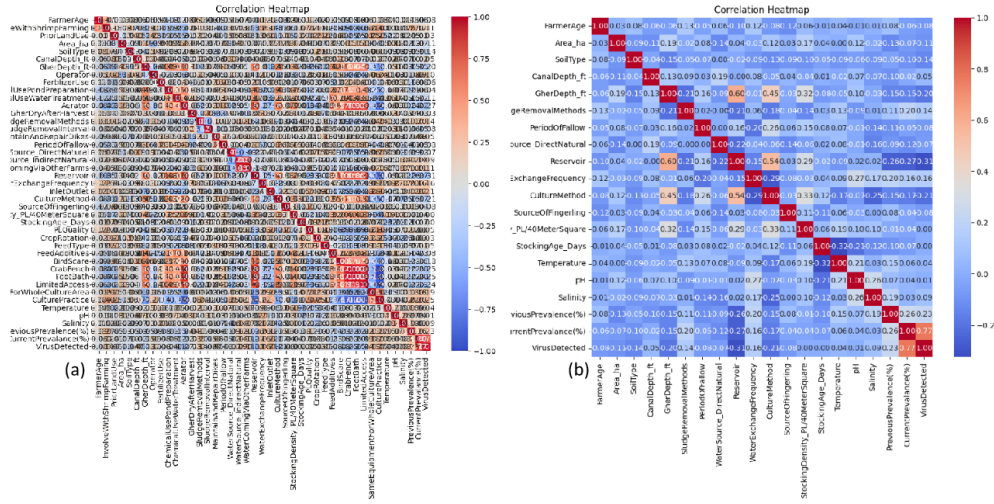


Fig. 2 Correlation Heat-map: (a) For all features (b) For the selected features by importance

Figure 3(a) and Figure 3(b) show the important features identified by the Random Forest and Multinomial Naive Bayes algorithm, respectively. From the figures, it can be seen that Multinomial Naive Bayes identified less number of features as important features than the Random Forest algorithm.

3.3 Machine Learning Model Selection

Once the feature engineering is done, the machine learning model is selected for the WSD prediction task. To select the appropriate model, each one is tested once

of 20, 5-fold cross-validation, a random state of 42, and verbosity set to 2 is an automated machine learning tool that evolves and optimizes a pipeline of classifiers using genetic programming, making the process more transparent (verbosity=2). Bagging with a Decision Tree Classifier using 10 estimators, a maximum of 80% of samples per estimator ($max - samples = 0.8$), and a random state of 42 is an ensemble technique that combines multiple decision trees for improved prediction stability and accuracy.

3.4 Evaluation Metrics

Evaluating the performance of prediction models is essential in understanding their effectiveness. Various metrics provide insights into different aspects of model performance. In the context of classification tasks, commonly used metrics include Accuracy (Acc), Precision ($Prec$), Recall (Rec), and the F1-score. Accuracy measures the overall correctness of predictions, while Precision and Recall focus on positive predictions and handling imbalanced datasets, respectively. The F1-score strikes a balance between Precision and Recall. Additionally, cross-validation scores, such as those obtained through k-fold cross-validation, offer a robust estimate of a model’s performance by considering multiple data splits. These metrics collectively contribute to a comprehensive evaluation of predictive models.

The equations for the metrics are as follows:

$$Acc = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$Prec = \frac{TP}{TP + FP} \quad (2)$$

$$Rec = \frac{TP}{TP + FN} \quad (3)$$

$$F1 = \frac{2 \cdot Prec \cdot Rec}{Prec + Rec} \quad (4)$$

Here, TP refers to True Positives, TN to True Negatives, FP to False Positives, and FN to False Negatives.

3.5 Experimental setup

In this study, we utilized Google Colab, a cloud-based platform, to design, implement, and train our machine learning models. Google Colab’s free version provided a powerful and accessible computing environment, enabling us to conduct extensive experiments without significant computational resources. The collaborative nature of Google Colab allowed seamless sharing of code and datasets among research team members, facilitating collaboration and improving research efficiency. All the codes were written in Python3, and Scikit Learn is used as the ML library.

4 Results and Discussion

This study employed various models with different combinations to investigate predictive modeling and early detection of White Spot Disease (WSD) in shrimp farming,

using a case study in Bangladesh. Random Forest and Logistic Regression models were implemented without feature selection, utilizing all features for model training. Table 2 presents the results of this experiment.

Table 2 Experimental results achieved from utilizing all the available features in the dataset in model training

Model	Cross Validation Score	Precision	Recall	F1-Score	Accuracy in %	AUC
Random Forest	0.978	1.0	0.968	0.964	97.87	1.00
Logistic Regression	0.94	0.96	0.84	0.89	87.23	0.72

Two models, Random Forest and Logistic Regression, were employed for WSD detection, with Logistic Regression estimating event probabilities using a dataset of independent variables [11]. The experiment involved using full features with outliers, where Random Forest outperformed Logistic Regression. After removing outliers and selecting features based on importance scores [12], three models were evaluated, and Table 3 displays the results.

Table 3 Experimental results achieved after outlier removal and feature selection

Model	Cross Validation Score	Precision	Recall	F1-Score	Accuracy in %	AUC
Random Forest	0.9784	1.0	0.968	0.984	97.87	0.99
Logistic Regression	0.905	0.96	0.84	0.89	87.23	0.93
Multinomial Naive Bayes	0.9784	0.98	0.97	0.97	97.87	0.99

Three distinct models, Random Forest, Logistic Regression, and Multinomial Naive Bayes, were used for WSD detection after removing outliers and unimportant features. Multinomial Naive Bayes exhibited superior performance, as shown in Table 3. The experiment with ensemble models and selected features is presented in Table 4.

Table 4 Experimental results achieved from ensemble models and selected features

Model	Precision	Recall	F1-Score	Accuracy in %	AUC
Random Forest with Decision Tree classifier	0.968	0.968	0.968	95.74	0.95
TPOT classifier	1.0	0.875	0.933	91.48	0.97
Bagging with Decision Tree classifier	1.0	0.968	0.984	97.87	0.98

Three ensemble models were examined for WSD detection, with Bagging using a Decision Tree Classifier demonstrating significantly superior performance. The results, obtained by removing unimportant features for ensemble models, highlight Bagging as the most effective strategy in terms of accuracy (Table 4).

Table 2 showcases the effectiveness of two models utilizing all available features, with the Random Forest classifier demonstrating remarkable precision in predicting

white spot disease at 97.87% accuracy and an AUC of 1.00. Table 3 presents three models that improve accuracy by removing outliers and unimportant features, with the Multinomial Naive Bayes model outperforming others, achieving an accuracy of 97.87% and an AUC of 0.99. In Table 4, three ensemble models focusing solely on removing unimportant features reveal the Bagging approach with the Decision Tree classifier as the most effective, attaining an accuracy of 97.87% and AUC of 0.98.

In a rigorous evaluation of different machine learning models for white spot disease detection, Random Forest and Bagging with Decision Trees exhibited identical overall predictive capabilities. The key differentiator emerged from the AUC score analysis, where Random Forest demonstrated a superior AUC score of 0.99 compared to Bagging with Decision Trees' 0.98. The nuanced difference suggests Random Forest's slightly enhanced discriminatory power, making it the preferred choice for this disease detection task.

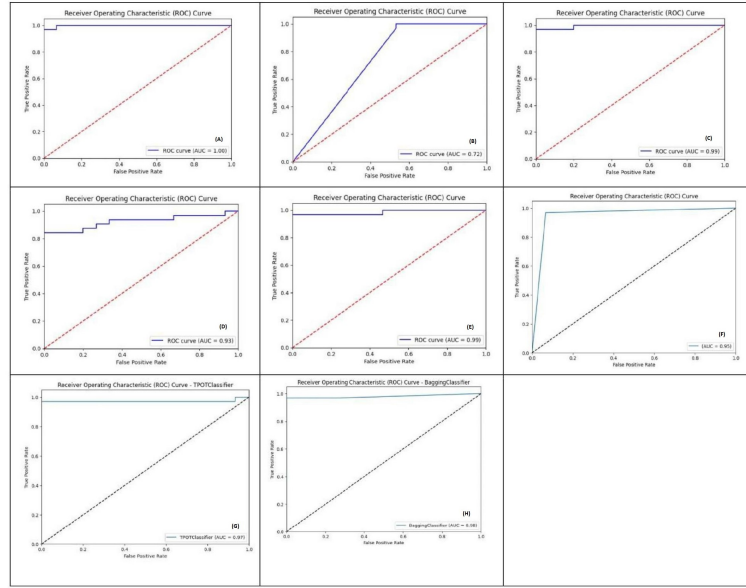


Fig. 4 ROC Curves for all the models employed in this study.

Figure 4 displays ROC-AUC curves, evaluating different models with varying subsets of data for enhanced WSD detection accuracy. Plots 4(a) and 4(b) show the results with comprehensive data utilization. Plot 4(a) with the Random Forest algorithm achieves a remarkable AUC score of 1.00, while plot 4(b) with Logistic Regression demonstrates comparatively inferior performance, yielding an AUC score of 0.72.

Plots 4(c), 4(d), and 4(e) depict outcomes of models designed to improve accuracy by eliminating outliers and unimportant features. Plot 4(e) with Multinomial Naive Bayes excels, attaining a robust AUC score of 0.99. Plots 4(c) and 4(d) with the Random Forest algorithm achieve AUC scores of 0.99 and 0.93, respectively.

Plots 4(f), 4(g), and 4(h) focus on eliminating only irrelevant features. The Bagging approach with the Decision Tree classifier emerges as the most effective, achieving an AUC score of 0.98. Random Forest with Decision Tree and TPOT Classifier achieve AUC scores of 0.95 and 0.97, respectively, showcasing diverse model performance across tasks for enhanced WSD detection accuracy.

The paper "Bootstrapping Random Forest and CHAID for WSD Prediction" notes the removal of many important features [8], which may impact real-world applicability. In contrast, our research uses multiple models to create a more reliable prediction system, capturing different aspects of the data for enhanced robustness. Parameters such as pH levels, water levels, current prevalence, and previous prevalence were identified as crucial for shrimp farming. Our approach appears promising for practical use in shrimp farming.

5 Conclusion and Future Work

In a Bangladesh case study, this research delves into predictive modeling for early detection of White Spot Disease (WSD) in shrimp farming. Through thorough data analysis and various machine learning algorithms, valuable insights are gained. Feature engineering significantly enhances predictive model accuracy. Techniques for explaining model predictions are explored to build trust among farmers and stakeholders. As the fight against WSD is ongoing and the disease evolves, continuous updates and refinements of the models are necessary to adapt to changing conditions and new data. Implementing these insights can contribute to more sustainable and resilient shrimp production systems in Bangladesh and beyond.

Future research areas include the development of automated data collection and monitoring systems, potentially incorporating IoT devices and sensor networks for real-time disease surveillance. Advanced machine learning techniques like deep and reinforcement learning can capture dynamic interactions within shrimp farming environments. Additionally, exploring transfer learning methods to adapt models trained in specific regions for use in new geographical areas could enhance prediction accuracy.

Data and Code Availability

The codes and data employed in this study can be found in the following GitHub repository. <https://github.com/ShahriarAyon63/Shrimp-WSD-Virus-Detection>

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